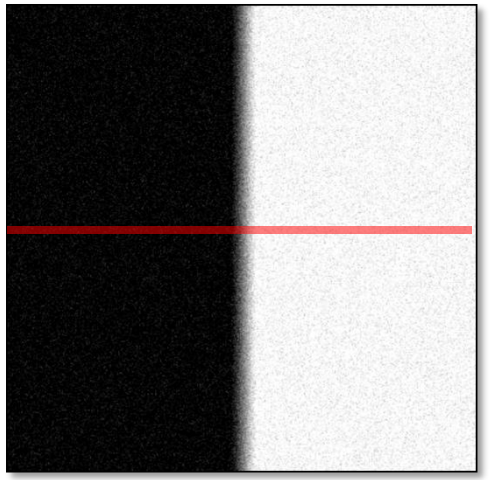
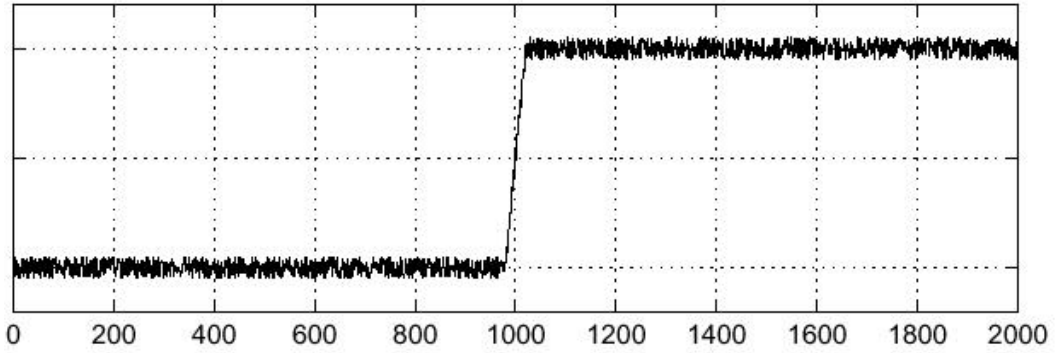
Effects of noise

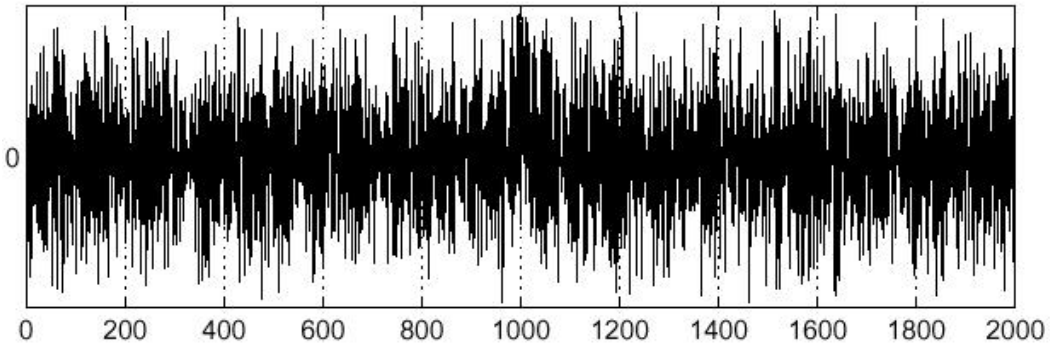


Noisy input image

$f(x)$



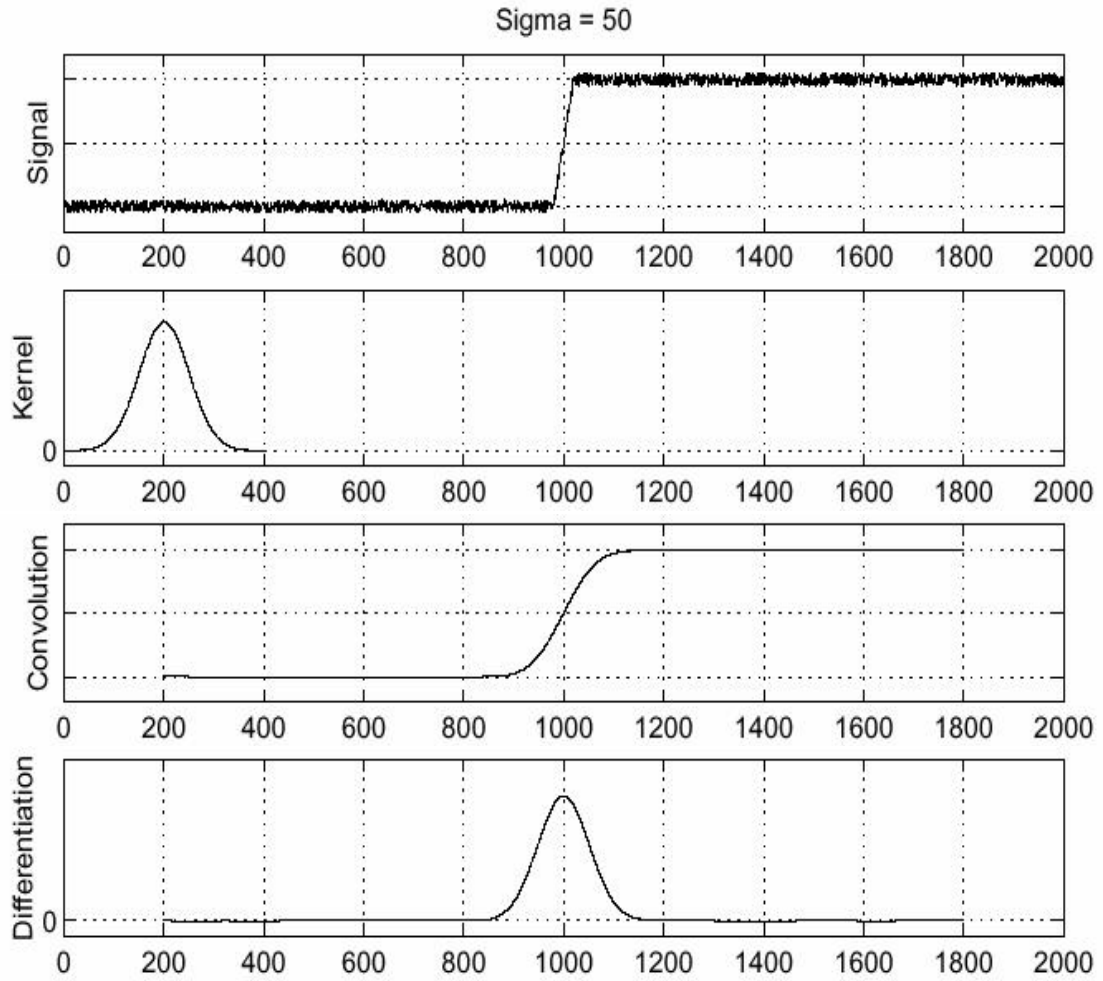
$\frac{d}{dx}f(x)$



Where is the edge?

Source: S. Seitz

Solution: smooth first



f

h

$f * h$

$\frac{d}{dx}(f * h)$

To find edges, look for peaks in $\frac{d}{dx}(f * h)$

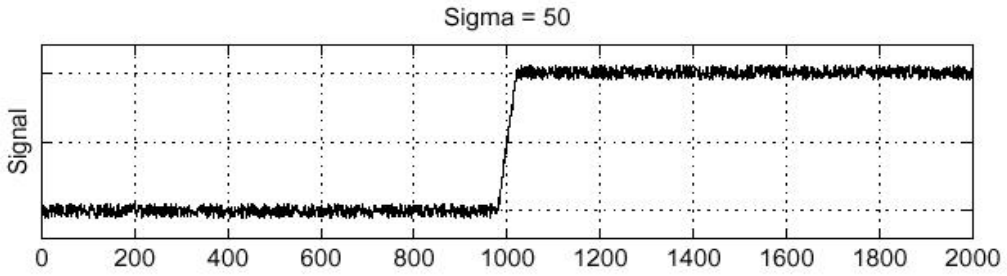
Source: S. Seitz

Associative property of convolution

- Differentiation is convolution, and convolution is associative:

$$\frac{d}{dx}(f * h) = f * \frac{d}{dx}h$$

f



$$\frac{d}{dx}(f * h) = \frac{d}{dx}\int f(t) \cdot h(x-t)dt = \int f(t) \cdot \frac{d}{dx}h(x-t)dt = f * \frac{d}{dx}h$$

- This saves us one opera

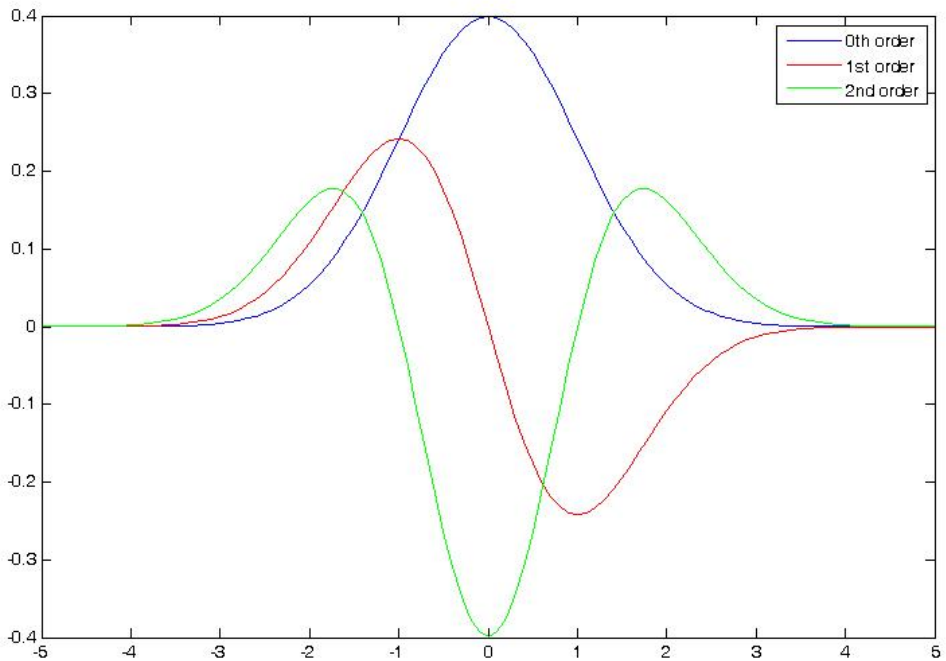
ax

Source: S. Seitz

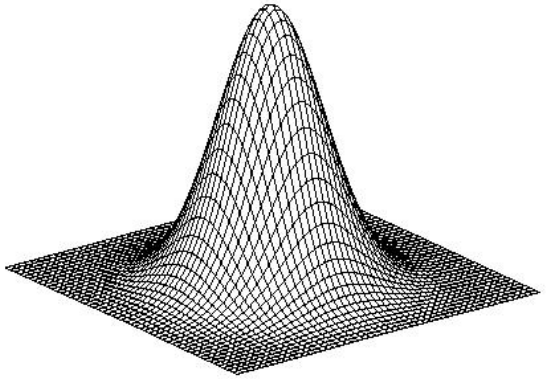
The 1D Gaussian and its derivatives

$$G_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

$$G'_\sigma(x) = \frac{d}{dx} G_\sigma(x) = -\frac{1}{\sigma}\left(\frac{x}{\sigma}\right) G_\sigma(x)$$

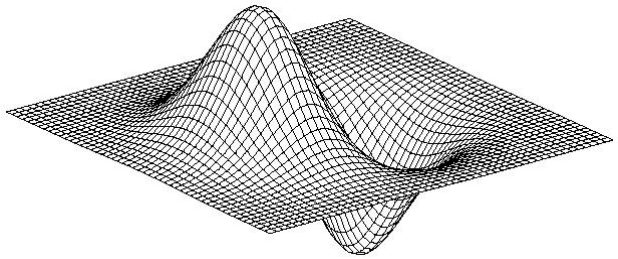


2D edge detection filters



Gaussian

$$h_\sigma(u,v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$

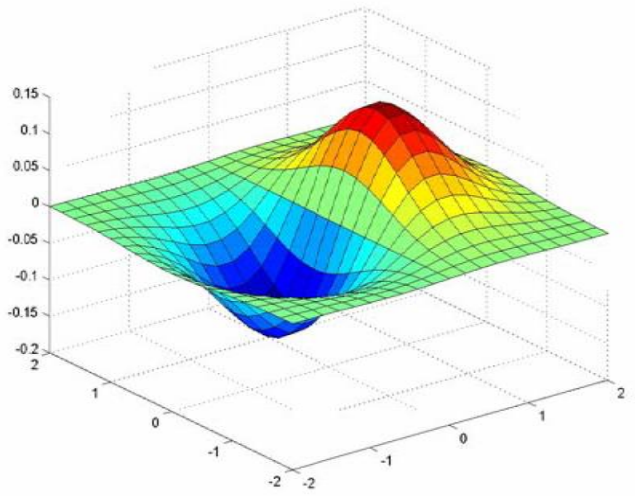


derivative of Gaussian (x)

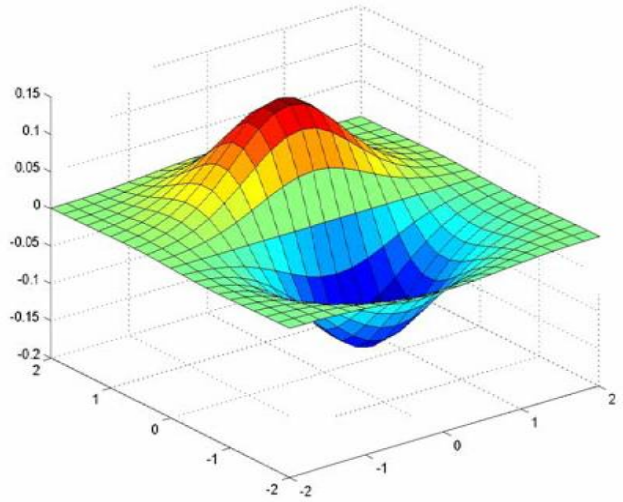
$$\frac{\partial}{\partial x} h_\sigma(u,v)$$

Derivative of Gaussian filter

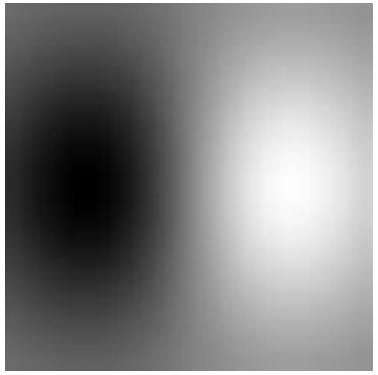
香港城市大學(東莞)
City University of Hong Kong
(Dongguan)

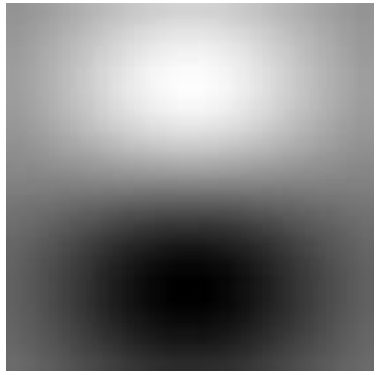


x-direction



y-direction





The Sobel operator

- Common approximation of derivative of Gaussian

$\frac{1}{8}$

-1	0	1
-2	0	2
-1	0	1

s_x

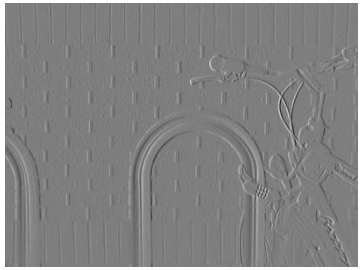
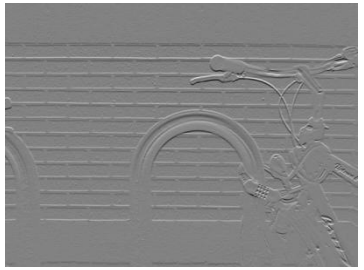
$\frac{1}{8}$

1	2	1
0	0	0
-1	-2	-1

s_y

- The standard definition of the Sobel operator omits the 1/8 term
  - doesn't make a difference for edge detection
  - the 1/8 term **is** needed to get the right gradient magnitude

Sobel operator: example




Source: Wikipedia